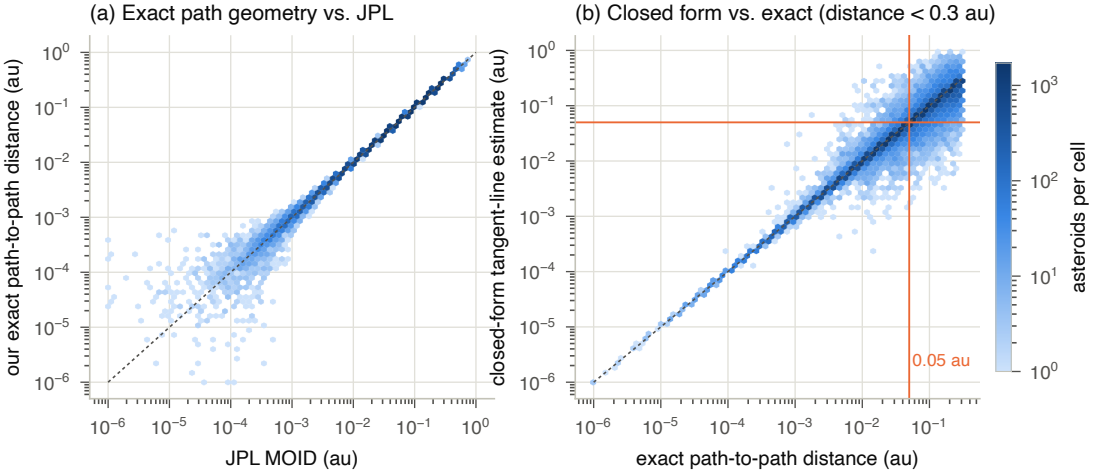

(a) Exact path geometry vs. JPL
our exact path-to-path distance (au)
JPL MOID (au)
(b) Closed form vs. exact (distance < 0.3 au)
closed-form tangent-line estimate (au)
exact path-to-path distance (au)
0.05 au
asteroids per cell